

Chapter 6: Conclusions and Future Work

6.1 Summary of findings

This thesis explored the feasibility of the streaming-causal BRAVE architecture running voice-driven instrument resynthesis with less than 10 ms latency when trained on heterogeneous, limited-scale datasets. The study went through six versions of the guitar model (v1–v6) and two versions of drums (drums_v1, drums_v2). It was found that guitar_v6, which ran on the official v2.gin and causal.gin configurations along with a smaller latent size of 16 and a longer Phase 1 of 1.5 M steps, delivered the first viable autoencoder for in-distribution input (Chapter 5). On the other hand, the drums_v2 version was stopped prematurely as the Phase 2 discriminator dominance appeared to have come back again even though an `update_discriminator_every=4` schedule had been implemented (Chapter 5).

It was concluded that voice remained an out-of-distribution (OOD) input all through the guitar checkpoints, which was the main goal of the cross-modal resynthesis (musical instruments from voice). The encoder seemed to distinguish voice inputs, with differences in latents observed ranging from 1.27 to 6.89, but the decoder output was almost the same all the time, signalling a structural failure in associating voice latents with the trained guitar manifold (Ch 5). Moreover, the OOD restriction happened to be due to the training recipe rather than a design mistake of BRAVE as per the comparison with the IRCAM percussion baseline. The external baseline was noticeably more reactive being 1.75 in forward difference whereas only 0.0045 was recorded in drums_v1 (roughly 390× greater responsiveness) and output given silent input was roughly 0.001 (Ch 5). Repeating the training iterations without stopping enabled the discovery of failures of RAVE-family models that operate with a low amount of data in five different aspects: (i) posterior collapse (version 1), (ii) latent underutilisation (version 2), (iii) discriminator domination or generator mode collapse (versions 3, 4, drums_v2), (iv) discriminator collapse (version 5), and (v) OOD decoder behaviour (versions 6, drums_v1) (Chapter 5).

6.2 Contributions

One of the major breakthroughs was recording the step-by-step method of an experiment that could uncover RAVE-family training failures. This stepwise approach includes comparing levels of checkpoints that respond to those that collapse, and through this, setting concrete levels as one of the ways to verify the model's internal representations (§3.4). Measuring differences between pairs both in the latent vectors and in the audio outputs allows one to do an exceptionally thorough evaluation that goes way beyond just doing informal listening tests.

The second major contribution is a failure taxonomy arranged in five modes. This paper interprets every TensorBoard fingerprint, for example, KL drift, logit gaps between real and fake, and output amplitude degeneration, and connects it to the corresponding programme. It is therefore recommended that this failure mode classification serve as a troubleshooting guide for those involved in training RAVE on small or heterogeneous data sets where training failure leading to loss of stability is a very frequent challenge (Ch 5).

A third significant point made in this thesis is that the system been benchmarked externally. The restricted reaction to OOD vocal inputs that were noticed in this study were due to the selection of the training data and hyperparameter settings rather than being an inherent weakness of the BRAVE architecture (Ch 5). To put it simply, if the training directions are further enhanced, the cross-modal discrepancy might be getting even closer to zero.

6.3 Limitations

Several limitations of this study should be considered. First of all, the latency, which was reported to be around 7 ms, was estimated analytically based on configuration parameters and reported buffer sizes (§3.1). This estimate was not corroborated by calibrated loopback measurements, which would be necessary to determine the actual end-to-end delay experienced in a real-world performance environment.

Secondly, the assessment of the audio quality and reactivity was done without using any formal auditory perceptual measures, e.g., MUSHRA (Multiple Stimuli with Hidden Reference and Anchor) or MOS (Mean Opinion Score). All listening tests were informal and performed by one author only, introducing bias and affecting the generalisability of the qualitative results (Ch 5).

Thirdly, data sources were quite limited in size for example, the guitar dataset contained only about 16 hours whereas the drum dataset about 11 hours which is considerably less than many community baselines for high fidelity synthesis (§3.3). Lastly, the voice evaluation was done using only one voice hence it is still unknown if the model is capable of generalizing to different speakers or different vocal characteristics.

6.4 Future work

Future research should involve calibrated loopback latency measurements. This testing will replace the current analytical estimates with real data, providing a definite confirmation of BRAVE's performance within the Pure Data environment and checking that the latency is within the bounds discussed by Wessel and Wright (2002) for intimate musical interaction.

Besides, a proper perceptual evaluation (e.g. MUSHRA/MOS) should be carried out to assess the in-distribution reconstruction quality of guitar_v6. This would make it possible to perform a statistically valid evaluation of the model's sound quality against original recordings and other real-time synthesis methods.

In the quest of producing better quality cross-domain resynthesis, the RAVE v3 architecture needs to be looked into. This involves Snake activations, a Descript style discriminator and Adaptive Instance Normalisation (AdaIN) for explicit style transfer. However, this kind of training would require hardware with at least 32 GB VRAM, whereas only 16 GB were used in this project (Ch 4).

The latent alignment could be tackled if paired or mixed-domain training data with both voice and instrument signals are used. It is assumed that giving the model exposure to both domains at the same time should not only lower the OOD distance but also foster the emergence of more responsive timbre transfer behaviour.

Even more importantly, there is huge scope in the creation of hybrid pipelines. Such systems, for example, could merge the use of symbolic onset detection and classification with a neural timbre transfer. The AVP dataset (Delgado et al., 2019), which contains annotated vocal percussion, may be employed to the development of a model capable of mapping different vocal phonemes to various drum sounds with high fidelity, far surpassing results achieved by a purely unsupervised autoencoder.

Finally, the validity of these findings has to be confirmed on a larger, more uniform corpus. Such a study will unveil if the IRCAM-style responsiveness is attributable only to the magnitude and uniformity of the data or whether additional architectural enhancements are necessary to minimize the disparity between human vocalisations and trained instrument domains across modalities.

6.5 Closing reflection

The original goal of this project was to build a voice-to-guitar working prototype; nevertheless, halfway through, it was changed to a documented case-study and failure-analysis project (Ch 5). This change of plan happened as it became apparent that although clear in-distribution reconstruction was attained, control of expressive OOD via human voice was not realised given the training conditions tested. The final effort was geared towards providing a diagnostic framework that can be reused and a systematic failure taxonomy rather than an unverifiable demonstration. The established framework here, it is suggested, will give a stronger basis for real-time neural synthesis research than an isolated successful demo could provide.